

MPG No. 8.4.27 $m'm'm$ (m'm'm setting) [Type III, orthorhombic] [T tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} 0 & T_{xy} & 0 \\ T_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{xy} = \mathbb{T}_{xy}^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & T_{xy} & 0 \\ -T_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{xy} = \mathbb{M}_z^{(1)}$$

* Rank 3 tensor (ss). * Rank 3 tensor (sa). * Rank 3 tensor (a). * Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & T_{xxxx} \\ 0 & 0 & 0 & 0 & 0 & T_{yyxy} \\ 0 & 0 & 0 & 0 & 0 & T_{zzxy} \\ 0 & 0 & 0 & 0 & T_{yzxz} & 0 \\ 0 & 0 & 0 & T_{yzxz} & 0 & 0 \\ T_{xxxx} & T_{yyxy} & T_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxxx} = \mathbb{T}_{4z}^{\alpha(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} - \mathbb{T}_z^{\beta(1)}$$

$$T_{yyxy} = -\mathbb{T}_{4z}^{\alpha(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} - \mathbb{T}_z^{\beta(1)}$$

$$T_{zzxy} = \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_z^{\beta(1)}$$

$$T_{yzxz} = \mathbb{T}_{xy}^{(2)} + 2\mathbb{T}_z^{\beta(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & T_{xxxx} \\ 0 & 0 & 0 & 0 & 0 & T_{yyxy} \\ 0 & 0 & 0 & 0 & 0 & T_{zzxy} \\ 0 & 0 & 0 & 0 & T_{yzxz} & 0 \\ 0 & 0 & 0 & -T_{yzxz} & 0 & 0 \\ -T_{xxxx} & -T_{yyxy} & -T_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxxx} = 2\mathbb{M}_z^{(1)} - 2\mathbb{M}_z^{\alpha(1)} + 2\mathbb{M}_z^{\beta(1)} + \mathbb{T}_{xy}^{(3)}$$

$$T_{yyxy} = -2\mathbb{M}_z^{(1)} + 2\mathbb{M}_z^{\alpha(1)} + 2\mathbb{M}_z^{\beta(1)} + \mathbb{T}_{xy}^{(3)}$$

$$T_{zzxy} = -4\mathbb{M}_z^{\beta(1)} + \mathbb{T}_{xy}^{(3)}$$

$$T_{yzxz} = -\mathbb{M}_z^{(1)} - 4\mathbb{M}_z^{\alpha(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} 0 & T_{yzxz} & 0 \\ T_{yzxz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzxz} = 2\mathbb{T}_{xy}^{(4)}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & T_{yzxz} & 0 \\ -T_{yzxz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzzx} = -\mathbb{M}_z^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & T_{xxxy} \\ 0 & 0 & T_{yyxy} \\ 0 & 0 & T_{zzxy} \\ 0 & T_{yzzx} & 0 \\ T_{zxyz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxxy} = \mathbb{M}_z^{(3)} - \mathbb{M}_z^{\alpha(2)} + \mathbb{M}_z^{\beta(2)} - 2\mathbb{T}_{xy}^{(5)}$$

$$T_{yyxy} = \mathbb{M}_z^{(3)} - \mathbb{M}_z^{\alpha(2)} - \mathbb{M}_z^{\beta(2)} + 2\mathbb{T}_{xy}^{(5)}$$

$$T_{zzxy} = \mathbb{M}_z^{(3)} + 2\mathbb{M}_z^{(4)} + 2\mathbb{M}_z^{\alpha(2)}$$

$$T_{yzzx} = \mathbb{M}_z^{(4)} - \mathbb{M}_z^{\alpha(2)} - \mathbb{M}_z^{\beta(2)} - \mathbb{T}_{xy}^{(5)}$$

$$T_{zxyz} = \mathbb{M}_z^{(4)} - \mathbb{M}_z^{\alpha(2)} + \mathbb{M}_z^{\beta(2)} + \mathbb{T}_{xy}^{(5)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & T_{yzzx} & 0 \\ 0 & 0 & 0 & T_{zxyz} & 0 & 0 \\ T_{xyxx} & T_{xyyy} & T_{xyzx} & 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzzx} = \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} + \mathbb{M}_z^{\beta(3)} + \mathbb{T}_{xy}^{(6)}$$

$$T_{zxyz} = \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} - \mathbb{M}_z^{\beta(3)} - \mathbb{T}_{xy}^{(6)}$$

$$T_{xyxx} = \mathbb{M}_z^{(5)} - \mathbb{M}_z^{\alpha(3)} + \mathbb{M}_z^{\beta(3)} - 2\mathbb{T}_{xy}^{(6)}$$

$$T_{xyyy} = \mathbb{M}_z^{(5)} - \mathbb{M}_z^{\alpha(3)} - \mathbb{M}_z^{\beta(3)} + 2\mathbb{T}_{xy}^{(6)}$$

$$T_{xyzx} = \mathbb{M}_z^{(5)} + 2\mathbb{M}_z^{(6)} + 2\mathbb{M}_z^{\alpha(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & T_{xxxy} & 0 & 0 & T_{xxyx} \\ 0 & 0 & 0 & 0 & 0 & T_{yyxy} & 0 & 0 & T_{yyyx} \\ 0 & 0 & 0 & 0 & 0 & T_{zzxy} & 0 & 0 & T_{zzyx} \\ 0 & 0 & 0 & 0 & T_{yzzx} & 0 & 0 & T_{yzxz} & 0 \\ 0 & 0 & 0 & T_{zxyz} & 0 & 0 & T_{zxzy} & 0 & 0 \\ T_{xyxx} & T_{xyyy} & T_{xyzx} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxxy} = 2\mathbb{M}_z^{(1)} + \mathbb{M}_z^{(3)} - 2\mathbb{M}_z^{\alpha(1)} - \mathbb{M}_z^{\alpha(2)} + 2\mathbb{M}_z^{\beta(1)} + \mathbb{M}_z^{\beta(2)} + \mathbb{T}_{4z}^{\alpha(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} - 2\mathbb{T}_{xy}^{(5)} - \mathbb{T}_z^{\beta(1)}$$

$$T_{xxyx} = 2\mathbb{M}_z^{(1)} - \mathbb{M}_z^{(3)} - 2\mathbb{M}_z^{\alpha(1)} + \mathbb{M}_z^{\alpha(2)} + 2\mathbb{M}_z^{\beta(1)} - \mathbb{M}_z^{\beta(2)} + \mathbb{T}_{4z}^{\alpha(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} + 2\mathbb{T}_{xy}^{(5)} - \mathbb{T}_z^{\beta(1)}$$

$$T_{yyxy} = -2\mathbb{M}_z^{(1)} + \mathbb{M}_z^{(3)} + 2\mathbb{M}_z^{\alpha(1)} - \mathbb{M}_z^{\alpha(2)} + 2\mathbb{M}_z^{\beta(1)} - \mathbb{M}_z^{\beta(2)} - \mathbb{T}_{4z}^{\alpha(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} + 2\mathbb{T}_{xy}^{(5)} - \mathbb{T}_z^{\beta(1)}$$

$$T_{yyyx} = -2\mathbb{M}_z^{(1)} - \mathbb{M}_z^{(3)} + 2\mathbb{M}_z^{\alpha(1)} + \mathbb{M}_z^{\alpha(2)} + 2\mathbb{M}_z^{\beta(1)} + \mathbb{M}_z^{\beta(2)} - \mathbb{T}_{4z}^{\alpha(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} - 2\mathbb{T}_{xy}^{(5)} - \mathbb{T}_z^{\beta(1)}$$

$$T_{zzxy} = \mathbb{M}_z^{(3)} + 2\mathbb{M}_z^{(4)} + 2\mathbb{M}_z^{\alpha(2)} - 4\mathbb{M}_z^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + \mathbb{T}_{xy}^{(3)} + 2\mathbb{T}_z^{\beta(1)}$$

$$T_{zzyx} = -\mathbb{M}_z^{(3)} - 2\mathbb{M}_z^{(4)} - 2\mathbb{M}_z^{\alpha(2)} - 4\mathbb{M}_z^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + \mathbb{T}_{xy}^{(3)} + 2\mathbb{T}_z^{\beta(1)}$$

$$T_{yzzx} = -\mathbb{M}_z^{(1)} + \mathbb{M}_z^{(4)} - 4\mathbb{M}_z^{\alpha(1)} - \mathbb{M}_z^{\alpha(2)} - \mathbb{M}_z^{\beta(2)} + \mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(5)} + 2\mathbb{T}_z^{\beta(1)}$$

$$T_{yzxz} = -\mathbb{M}_z^{(1)} - \mathbb{M}_z^{(4)} - 4\mathbb{M}_z^{\alpha(1)} + \mathbb{M}_z^{\alpha(2)} + \mathbb{M}_z^{\beta(2)} + \mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(5)} + 2\mathbb{T}_z^{\beta(1)}$$

$$T_{zxyz} = \mathbb{M}_z^{(1)} + \mathbb{M}_z^{(4)} + 4\mathbb{M}_z^{\alpha(1)} - \mathbb{M}_z^{\alpha(2)} + \mathbb{M}_z^{\beta(2)} + \mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(5)} + 2\mathbb{T}_z^{\beta(1)}$$

$$T_{zxzy} = \mathbb{M}_z^{(1)} - \mathbb{M}_z^{(4)} + 4\mathbb{M}_z^{\alpha(1)} + \mathbb{M}_z^{\alpha(2)} - \mathbb{M}_z^{\beta(2)} + \mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(5)} + 2\mathbb{T}_z^{\beta(1)}$$

$$T_{xyxx} = -2\mathbb{M}_z^{(1)} + 2\mathbb{M}_z^{\alpha(1)} - 2\mathbb{M}_z^{\beta(1)} + \mathbb{T}_{4z}^{\alpha(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(3)} - \mathbb{T}_z^{\beta(1)}$$

$$T_{xyyy} = 2\mathbb{M}_z^{(1)} - 2\mathbb{M}_z^{\alpha(1)} - 2\mathbb{M}_z^{\beta(1)} - \mathbb{T}_{4z}^{\alpha(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(3)} - \mathbb{T}_z^{\beta(1)}$$

$$T_{xyzx} = 4\mathbb{M}_z^{\beta(1)} + \mathbb{T}_{xy}^{(1)} - \mathbb{T}_{xy}^{(3)} + 2\mathbb{T}_z^{\beta(1)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & T_{yzzx} & 0 & 0 & T_{yzxz} & 0 \\ 0 & 0 & 0 & T_{zxyz} & 0 & 0 & T_{zxzy} & 0 & 0 \\ T_{xyxx} & T_{xyyy} & T_{xyzx} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} T_{yzzx} &= -\mathbb{M}_z^{(2)} + \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} + \mathbb{M}_z^{\beta(3)} + 2\mathbb{T}_{xy}^{(4)} + \mathbb{T}_{xy}^{(6)} \\ T_{yzxz} &= \mathbb{M}_z^{(2)} + \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} + \mathbb{M}_z^{\beta(3)} - 2\mathbb{T}_{xy}^{(4)} + \mathbb{T}_{xy}^{(6)} \\ T_{zxyz} &= \mathbb{M}_z^{(2)} + \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} - \mathbb{M}_z^{\beta(3)} + 2\mathbb{T}_{xy}^{(4)} - \mathbb{T}_{xy}^{(6)} \\ T_{zxzy} &= -\mathbb{M}_z^{(2)} + \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} - \mathbb{M}_z^{\beta(3)} - 2\mathbb{T}_{xy}^{(4)} - \mathbb{T}_{xy}^{(6)} \\ T_{xyxx} &= \mathbb{M}_z^{(5)} - \mathbb{M}_z^{\alpha(3)} + \mathbb{M}_z^{\beta(3)} - 2\mathbb{T}_{xy}^{(6)} \\ T_{xyyy} &= \mathbb{M}_z^{(5)} - \mathbb{M}_z^{\alpha(3)} - \mathbb{M}_z^{\beta(3)} + 2\mathbb{T}_{xy}^{(6)} \\ T_{xyzx} &= \mathbb{M}_z^{(5)} + 2\mathbb{M}_z^{(6)} + 2\mathbb{M}_z^{\alpha(3)} \end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & T_{xxxy} & & 0 \\ T_{yyyyx} & 0 & & 0 \\ 0 & 0 & & 0 \\ 0 & 0 & & 0 \\ 0 & T_{zzxy} & & 0 \\ T_{xxyx} & 0 & & 0 \\ T_{yzzx} & 0 & & 0 \\ 0 & 0 & & 0 \\ 0 & T_{xyyy} & & 0 \\ 0 & 0 & \frac{T_{xxxy}}{2} - \frac{T_{xxyx}}{2} + T_{xyyy} - T_{yyyx} + T_{zzxy} & \end{bmatrix}$$

$$\begin{aligned} T_{xxxy} &= \mathbb{T}_{4z}^{\alpha(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} - 2\mathbb{T}_{xy}^{(5)} - \mathbb{T}_z^{\beta(1)} \\ T_{yyyyx} &= -\mathbb{T}_{4z}^{\alpha(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} - 2\mathbb{T}_{xy}^{(5)} - \mathbb{T}_z^{\beta(1)} \\ T_{zzxy} &= \mathbb{T}_{xy}^{(1)} + \mathbb{T}_{xy}^{(3)} + 2\mathbb{T}_z^{\beta(1)} \\ T_{xxyx} &= \mathbb{T}_{4z}^{\alpha(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} + 2\mathbb{T}_{xy}^{(5)} - \mathbb{T}_z^{\beta(1)} \\ T_{yzzx} &= \mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(5)} + 2\mathbb{T}_z^{\beta(1)} \\ T_{xyyy} &= -\mathbb{T}_{4z}^{\alpha(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(3)} - \mathbb{T}_z^{\beta(1)} \end{aligned}$$